Varun Tripathi

AI-Augmented QA Engineer | Test Automation & Intelligent Systems | Product-minded Builder
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EDUCATION

- **Indian Institute of Information Technology (IIIT)** *Nagpur, India*
  B.Tech in Electronics and Communication Engineering (IoT) 08/23 – 07/27

SKILLS

Languages: Python, TypeScript, SQL, JavaScript

AI & Machine Learning: LangGraph, CrewAI, PyTorch, Groq (Llama-3.3-70B), Claude API, Agentic Workflows, Tool Calling, Prompt Engineering, RAG

QA & Test Automation: Playwright, pytest, unittest, Unit Testing, Test Cases Design, API Testing, Test-Driven Development (TDD), Page Object Model, Self-Healing Locators, Visual Regression Testing, Defect Analysis, MCP Fuzzing, Flaky Test Analysis, CI/CD Pipelines

Retrieval & Data: FAISS, ChromaDB, Embeddings, Chunking, NumPy, Pandas, PostGIS

Backend & Cloud: FastAPI, PostgreSQL, Redis, Docker, AWS, WebSockets, Server-Sent Events, REST APIs, Git

Process: Agile, Quality Assurance, Bug Triage, Code Review, Modular Design

PROJECTS

- **MCP Test Framework (Playwright + Python + AI)** | GitHub *09/26 – 09/26*
  - Built an AI-augmented test automation framework for MCP servers on Playwright and pytest, shipping self-healing locators, visual regression triage, MCP fuzzing, flaky-test root-causing, and NL-to-test generation across 39 files and 1,270 lines.
  - Designed every AI feature behind a flag with graceful degradation – the core deterministic suite (unit, MCP, UI tests) runs green with zero AI dependency, while the AI modules plug in via the Claude API for semantic assertions and synthetic data generation.
  - Structured the framework with Page Object Model, pytest fixtures, async MCP client (stdio + SSE), per-environment config, and HTML reporting with screenshot capture – runnable end-to-end on a fresh clone against a bundled FastMCP demo server.
- **HireSignal (AI Hiring Intelligence Platform)** | GitHub *08/26 – 08/26*
  - Scored hiring momentum for 295 companies from 886 sources (676 HTTP, 132 RSS, 75 ATS API) deterministically, gating extraction behind a rule filter and an 8B classifier.
  - Engineered a 9-table pipeline on FastAPI, PostgreSQL/pgvector, Celery and Groq (Llama-3.1-8B) with a 6K context cap, a 0.30 confidence floor and 100:1 embedding batching.
  - Validated with 319 pytest tests passing in 14s at 0.96 mean confidence, classified 434/434 jobs with 0 LLM calls via the deterministic path, and cut model spend 12x.
- **Legal Contract Analyzer (RAG Backend System)** | GitHub *02/26 – 03/26*
  - Deployed a Python/FastAPI RAG API for 10MB contracts, integrating Groq LPUs (LLaMA-3) and Server-Sent Events (SSE) streaming to slash Time-To-First-Token from 26.7s to 0.2s (∼90x speedup).
  - Accelerated FAISS retrieval 40x (to 10ms) using local CPU embeddings (BAAI/bge-small), enforcing strict Pydantic JSON schemas to mitigate LLM hallucinations.
  - Managed multi-turn Q&A via a Redis distributed session store using ZSET LRU eviction, capping concurrent traffic at 250 users while cutting document processing time by 4x.
- **DevAgent Remote – Mobile Control Plane for AI Coding Agents** *07/26 – 07/26*
  - Built a control plane driving CLI coding agents (Claude Code, Gemini CLI, Aider) on a laptop from a mobile client over an outbound WebSocket tunnel, exposing 0 inbound ports and Git-checkpointing every task for rollback.
  - Decoupled the system into a FastAPI relay, an asyncio worker and an Expo React Native client behind a 13-message JSON protocol secured with GitHub OAuth 2.0 and JWT.
  - Delivered 4,508 lines across 31 files in 10 days (91 commits), integrating 3 CLI agent adapters with 1s-to-30s exponential reconnect backoff for fault tolerance.

LEADERSHIP

- **E-Cell, IIIT Nagpur** *Lead*
  Secured INR 11L+ from 10+ sponsors; directed 20+ volunteers across 17 events, 2,500+ attendees. 08/23 – 04/25
- **TantraFiesta Tech Fest** *Marketing Lead*
  Grew turnout 16.7% YoY to 1,400+ participants and 5,000+ crowd through measured campaigns. 08/24 – 10/24

ACHIEVEMENTS

- **1st Place, Maximise (E-Summit 2026):** Highest ROI on a INR 10,00,000 virtual portfolio in a 10-day stock trading simulation.
- **1st Rank, Ad Mads (Abhivyakti 2026):** First place in a nationwide creative campaign competition.
- **LeetCode Knight:** Rating 1886, 300+ problems solved | 30+ public GitHub repositories.